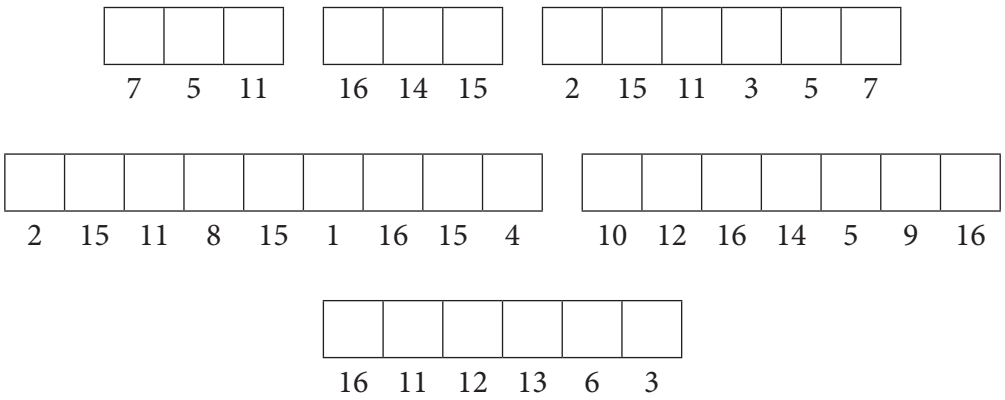
WORKSHEET

Chapter 3: Inverse functions

A diamond cannot be polished without friction …



Chinese proverb

Questions

Find the inverse functions of the following.

1 $f(x) = 5x + 3$

2 $g(x) = 2x - 1$

3 $h(x) = x^2$

4 $f(x) = \frac{1}{4}x^2 - 3$

5 $g(x) = 7x + 5$

6 $h(x) = \frac{1}{3}x - 2$

7 $f(x) = 3x^2 - 8$

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8 $g(x) = (x + 4)^2$

9 $h(x) = 2x^2 + 9$

10 $f(x) = \left(\frac{1}{5}x - 3\right)^2 + 1$

Given $f(x) = 4x + 7$, find the following, correct to two decimal places if necessary:

11 $f^{-1}(x)$

12 $f^{-1}(2)$

13 $f^{-1}(-1)$

Given $g(x) = \frac{1}{2}x^2 - 3$, find the following, correct to two decimal places if necessary:

14 $g^{-1}(x)$

15 $g^{-1}(-3)$

16 $g^{-1}(0)$

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Solutions

A -2

C $\dfrac{x-3}{5}$

D $\sqrt{4(x+3)}$

E 0

F $\sqrt{x}-4$

H $\sqrt{2(x+3)}$

I $-1\frac{1}{4}$

L $3(x+2)$

N $\sqrt{\dfrac{x+8}{3}}$

O $\dfrac{x-5}{7}$

P $\dfrac{x+1}{2}$

R $\dfrac{x-7}{4}$

S $\sqrt{x}$

T 2.45

U $\sqrt{\dfrac{x-9}{2}}$

W $5(\sqrt{x-1}+3)$

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